SELECTED AI & AUTOMATION PROJECTS

Ho Yeun Jae | +82 10-5235-6597 | hominid3142@gmail.com

AI Solutions Technician — Nexovo Solutions Inc.
Portfolio: https://github.com/Molut/ai-automation-portfolio

I define requirements, reproduce failures and direct coding agents through implementation and verification. The projects below combine AI integrations with practical school and scheduling workflows.

1. LUXID | AI messenger and workflow application

Independent project • Node.js, PostgreSQL, model APIs, browser automation, TTS

Developed with coding agents a messenger that combines configurable AI responses, long-term memory, selected conversation context, browser research and voice/image features. The main challenge was making these services work reliably together rather than treating each model call as an isolated demo.

- Reliable execution: separated reply generation from the browser request using durable jobs, request keys and leases. Repeated API submissions stored one user message and one reply; a real PostgreSQL restart test checked recovery and duplicate prevention.
- Multimodal integration: repaired case-sensitive multipart parsing, passed image interpretation into the existing chat model, and resolved an FFmpeg frame-padding failure in multi-speaker audio assembly. A subsequent 24-turn call completed as a 129.894-second MP3.
- Browser lifecycle: replaced full-profile copying with a minimal authenticated session copy and ownership-scoped cleanup. Three repeated live runs left no temporary profile directories; lifecycle regression checks covered timeout and late-completion cases.

2. LINCOLN SCHOOL | Repeatable teaching and administrative workflows

Workplace automation • Python, Google Sheets/Drive, XLSX, DOCX, browser automation

Connected monthly lesson plans, actual teaching history and class-specific rules to daily schedules and teaching logs. Replaced repeatable LLM-driven steps with deterministic Python scripts while retaining agent assistance for exceptions and maintenance.

- Handled skipped lesson-history records and separate Friday classes, restricted spreadsheet writes to the intended cells, and read back the updated values. Added dated CSV records to support later comparison and correction.
- Extended the workflow to document generation and vocabulary-test preparation, using the selected lesson/unit as the source rather than re-entering the same information across outputs.

These projects show both sides of the role: integrating AI services where judgment is useful, and using ordinary code where predictable execution is the better solution.

SELECTED PROJECTS — CONTINUED

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3. HAEJO | AI automation-assessment web app

Independent project • AI recommendations, Node/Express, Linux services, reverse proxy

Developed an assessment flow that presents automation-use-case recommendations and an AI diagnosis. Worked with coding agents on product behavior, API integration and deployment.

- Output contract: traced an eight-item UI request returning four items to a backend batch limit. Aligned the API limit, added handling for short model output and checked the eight-item response after deployment. Deterministic fallback entries could supplement short model output.
- Service recovery: investigated a public 502 while the application process was still alive. Isolated a missing proxy dependency, restored the Linux service and checked the public response rather than relying on localhost health alone.

4. ASSIGNMENT PLANNER | Scheduling decisions to official documents

Volunteer operations • JavaScript PWA, Python API, Excel, PDF forms, Google Drive

Directed development of a scheduling interface around the assignment-summary source sheet, showing role-specific candidates, recent workload, assignment intervals and same-week duplicate warnings. Connected edits back to Excel and generated individual assignment documents.

- Source integrity: kept the assignment-summary sheet separate from formula-driven monthly views; checked range-level write-back against the workbook and refreshed generated app data when the source changed.
- Document accuracy: filled the official PDF template rather than recreating it. A verified batch produced 16 PDF/PNG pairs, zero source-to-PDF field mismatches and 32 matching files in the shared Drive output.

SUPPORTING AGENT & SYSTEM OPERATIONS

- MCP integration: connected Hermes to REA/Ghidra and verified initialization, discovery of 170 tools, a real test-binary analysis and session closure.
- Linux and service diagnostics: established SSH-key access to Ubuntu and used systemd/journal logs to distinguish gateway, synchronization-path and network-authentication failures.
- Authentication and LLM operations: worked with OAuth refresh failures, organization OWA mail integration, hosted-model access and local LLM API/lifecycle checks.

HOW I WORK

Start from the actual task and source data; use AI agents for implementation and investigation; reproduce the failure; verify the API, saved state and final user-visible result; document the repeatable path. My translation and interpreting background supports clear requirements, careful terminology and practical technical documentation.